

Job-Order Costing, Over/Under-Applied Overhead & ABC —

Boulangerie Saint-Louis

ACCT 2102 | Chapter 6 in-class problem (full job-cost flow + activity-based costing) | ~25 min

Boulangerie Saint-Louis is a wholesale bakery in Metz that **manufactures baked goods to order** for local cafés, brasseries, and grocers. It uses a **job-order costing system** and applies manufacturing overhead with a **plantwide predetermined rate of \$60 per direct labor hour**, set from estimates at the start of the year. Direct labor is paid **\$20 per hour**.

The four jobs in the shop during the week of 15 June

Job	Product	Quantity	Status at the end of the week
100	Pâté lorrain	200	Was already in process at the start of the week; finished and delivered to the customer
101	Pain au chocolat	96 (8 doz)	Started, finished, and delivered this week
102	Baguette	400	Started and finished this week; in the cooler awaiting pickup
103	Quiche lorraine	150	Started this week; still in process at week's end

Job 100 was **already in process** at the start of the week with **\$300 of cost accumulated** (beginning Work in Process). There was no beginning Finished Goods inventory.

Direct costs added this week

Job	Direct materials	Labor hours	Direct labor
100 Pâté lorrain	\$40	4	\$80
101 Pain au chocolat	\$120	12	\$240
102 Baguette	\$160	8	\$160
103 Quiche lorraine	\$180	6	\$120
Total	\$500	30	\$600

Actual manufacturing overhead incurred this week

Overhead cost	Amount
Oven gas & utilities	\$520
Indirect labor (production manager, cleaner)	\$700
Equipment depreciation	\$300
Indirect materials (packaging, parchment)	\$180
Factory rent & insurance	\$250
Total actual overhead	\$1,950

Activity data (for the ABC part)

Management identified three activity cost pools with the following predetermined rates, and recorded each job's actual usage:

Activity (rate)	Job 100	Job 101	Job 102	Job 103
Mixing & prep — \$20/labor-hr	4 hr	12 hr	8 hr	6 hr
Baking — \$30/oven-load	4 loads	12 loads	3 loads	5 loads
Setup & packaging — \$40/run	3 runs	6 runs	1 run	2 runs

Required

(a) Job cost sheets (plantwide). For each job, accumulate beginning WIP + direct materials + direct labor + overhead applied at \$60/labor-hour, to get total cost (and cost per unit).

(b) Account balances. Based on each job's status, determine the ending balance in **Work in Process** and **Finished Goods**, and the **unadjusted Cost of Goods Sold**.

(c) Over- or under-applied overhead. Compare overhead applied to actual overhead. Is overhead over- or under-applied, and by how much? Give the entry to close it to COGS and the **adjusted COGS**.

(d) ABC overlay. Recompute each job's overhead using the three activity pools. Which jobs did the plantwide rate *undercost* and which did it *overcost*? What does switching to ABC do to total over/under-applied overhead, and to the margins on the jobs that were sold?

Solution Key

Full worked solution.

(a) Job cost sheets — plantwide (\$60 per labor hour)

Overhead applied = labor hours × \$60. Total cost = beginning WIP + DM + DL + OH applied.

Job	Beg. WIP	DM	DL	OH applied	Total	Per unit
100 Pâté lorrain	\$300	\$40	\$80	\$240	\$660	\$3.30
101 Pain au chocolat	—	\$120	\$240	\$720	\$1,080	\$11.25
102 Baguette	—	\$160	\$160	\$480	\$800	\$2.00
103 Quiche lorraine	—	\$180	\$120	\$360	\$660	\$4.40
Total	\$300	\$500	\$600	\$1,800	\$3,200	—

OH applied: $4 \times 60 = \$240$; $12 \times 60 = \$720$; $8 \times 60 = \$480$; $6 \times 60 = \$360 \rightarrow$ **\$1,800 total applied**. Per unit = total ÷ quantity (e.g., $\$1,080/96 = \11.25).

(b) Account balances from job status

Completed this week = Jobs 100, 101, 102 (cost of goods manufactured = $\$660 + \$1,080 + \$800 =$ **\$2,540** transferred from WIP to Finished Goods). Of those, Jobs 100 and 101 were **sold**.

Account	Contents	Balance
Ending Work in Process	Job 103 (still in process)	\$660
Ending Finished Goods	Job 102 (finished, not sold)	\$800
Unadjusted Cost of Goods Sold	Jobs 100 + 101 (sold)	\$1,740

Check: costs to account for = beg WIP \$300 + DM \$500 + DL \$600 + OH applied \$1,800 = **\$3,200** = ending WIP \$660 + ending FG \$800 + unadjusted COGS \$1,740. Ties out.

(c) Over- or under-applied overhead

	Amount
Overhead applied (from part a)	\$1,800
Actual overhead incurred	\$1,950
Under-applied overhead (actual > applied)	\$150

Applied < actual, so overhead is **under-applied by \$150** — not enough cost was pushed into the jobs, so **COGS must increase**. Closing entry (amount is immaterial → close straight to COGS):

Dr Cost of Goods Sold \$150 Cr Manufacturing Overhead \$150

Adjusted COGS = \$1,740 + \$150 = \$1,890. (Ending WIP and Finished Goods are unchanged.)

(d) ABC overlay — apply the three activity rates to each job

Job	Mixing \$20/hr	Baking \$30/load	Setup \$40/run	ABC OH	Plantwide
100 Pâté lorrain	$4 \times 20 = \$80$	$4 \times 30 = \$120$	$3 \times 40 = \$120$	\$320	\$240

Job	Mixing \$20/hr	Baking \$30/load	Setup \$40/run	ABC OH	Plantwide
101 Pain au choc.	12×20 = \$240	12×30 = \$360	6×40 = \$240	\$840	\$720
102 Baguette	8×20 = \$160	3×30 = \$90	1×40 = \$40	\$290	\$480
103 Quiche	6×20 = \$120	5×30 = \$150	2×40 = \$80	\$350	\$360
Total	\$600	\$720	\$480	\$1,800	\$1,800

- **Plantwide undercosts** the labor-light but oven-/setup-heavy jobs: pain au chocolat (\$720 → \$840, +\$120) and pâté lorrain (\$240 → \$320, +\$80). **It overcosts** the simple, high-volume baguette job (\$480 → \$290, −\$190). Quiche is about right (−\$10). The reason: a labor-only rate ignores that baking and setup are driven by oven-loads and production runs, which the small specialty batches consume heavily relative to their labor.
- **Total applied is still \$1,800**, so switching to ABC **does not change** total over/under-applied overhead (\$150 under-applied either way) — ABC changes *which job* absorbs cost, not the total.
- **Margin punchline:** the two jobs that were sold (100 and 101) are exactly the ones the plantwide rate *undercosts*. Under ABC their COGS would be \$320 + \$840 = \$1,160 instead of \$240 + \$720 = \$960 — \$200 higher. The plantwide rate flatters the reported margin on the very orders that left the building, while parking too much cost in the unsold baguette inventory.